

# APPENDIX PMD — POSITION–MASS DUALITY

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## The Context-Price of a Definite Location — how *localization* is bought with **mass** as well as with **space**

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*"To be anywhere in particular, a thing must pay twice: once in space, by giving up everywhere; and once in mass, by letting the rest of the universe hold it in place. A 'here' is context that has begun to weigh."*

— Petar Nikolov, opening the PMD hypothesis, 2026-07-19

*(Figurative framing. The defensible core claim is **not** gravitational weight or a literal Machian mechanism — it is §30's Localization-Leverage: mass sets localization **capacity**, context realizes **access**. The Machian reading is flagged open throughout, §4 / §8 / §30.)*

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**Framework:** U-Theory v31 · deepens the **Position** ↔ **Space** currency

**Status:** L3 / L4 SPECULATIVE (Part I) + L1/L2/L3 FORMAL (Part II) — a coherent interpretive extension of the canon, **NOT** a derivation of new physics, **NOT** a modification of GR or QFT. (v1.5.1 — mathematical-apparatus upgrade, adversarially corrected: **PMD-2R** exact relativistic bound (closes the NR scope gap) + the parametrization-reversal caveat; **PMD-10** capacity–efficiency factorization (removes the mass double-count in the adapter); **PMD-11** reference-scale invariance — *proved per-meter, disproved for the combined score*, with the dominance / trade-off dichotomy; **PMD-12** exact surplus identity  $\Delta \log\text{-loss} = I(Y;C_R|X_{\text{std}})$ ; plus one concrete falsifiable ceiling. **v1.5.1** corrects three errors found by hostile review of the v1.5 draft — see the changelog; the errors and their repairs are recorded, not silently removed.)

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**Prerequisites:** the core triad  $\text{Form} \leftrightarrow \text{Time} \cdot \text{Position} \leftrightarrow \text{Space} \cdot \text{Action} \leftrightarrow \text{Energy}$ ;

**APPENDIX\_MMT** (Meaning–Matter Transformation), **APPENDIX\_DIM** (Dimensionless Meaning), **APPENDIX\_ST** / DPR (Dimensional Price Registry), **APPENDIX\_QMC** (measurement collapse), **APPENDIX\_GEN** (Genesis Law), **APPENDIX\_SSS** (the score); for Part II, basic quantum estimation theory (Fisher information, Cramér–Rao, quantum Fisher information) and **APPENDIX\_RH** (null-model / rival discipline).

**How to read this document. PART I (§0–§8)** is the *interpretation*: what the Position↔Mass duality means, its honest scope, and where it is speculative (L3/L4). **PART II (§9–§30)** is the *formal layer*: what can actually be **proved** (identities and information inequalities, L1/L2), what is a **declared modelling choice** (the  $[0,1]$  meters, L3), and what is **forbidden** without a new observable (the no-go theorem). If you want anything quantitative, go to Part II. **Minimal reading paths**: *conceptual reader* — Part I only, stop at §8; *quantitative user* — §12 (the bound), §17 (the adapter), §23 (the no-go), §30 (the core claim).

# PART I — INTERPRETATION

## 0. What this appendix IS and IS NOT

| This appendix IS                                                                                          | This appendix IS NOT                                                        |
|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| A sharpening of the <b>Position</b> currency: a location has <i>two</i> non-substitutable prices, not one | A new ontology or a new fundamental theory                                  |
| A reading of the real, textbook link between <b>mass and localizability</b> through the F/P/A lens        | A derivation from the L1 axioms                                             |
| Anchored on established physics where it is established (Compton wavelength, inertia)                     | A claim to overturn or replace GR / QFT                                     |
| Explicit about which step is rigorous and which is speculative interpretation                             | A claim of empirical proof of "context = mass"                              |
| Marked <b>L3/L4 speculative</b> throughout, with honest failure hooks (§8)                                | A claim that a delocalized state <i>must</i> be massless (it need not — §8) |

**Core thesis (domain-limited).** *Position* ↔ *Space* is the canon. PMD adds a **narrower** bridge: **for a massive one-particle state in a chosen rest-frame description**, invariant mass fixes the reduced-Compton scale  $\lambda_C = \hbar/(mc)$  at which a fixed-particle-number localization becomes QFT-sensitive, and it sets the **inertial response to changes in momentum**. U-Theory reads these two roles as a **mass-sensitive sub-meter within the Position pillar** — one proxy for localizability, not a universal one. **Massless field excitations (photons) need a separate localization/detection meter and are NOT assigned zero Position merely because their invariant mass vanishes** (§5, §8).

## 1. The question (stated cleanly)

The canon says **resistance-against-Form = Time**, **resistance-against-Position = Space**, **resistance-against-Action = Energy**. Position is priced in Space: to occupy a unique location you must **exclude** a region — you give up being everywhere.

But a location has a *second* face the canon leaves implicit:

- For a **massive** particle, a sharp rest-frame "**here**" is bounded below by the reduced-Compton scale  $\lambda_C = \hbar/(mc)$ , so mass **sets the rest-frame localization scale** (*it does not follow that massless quanta cannot be localized at all* — §8).
- To **change its state of motion**, a system must overcome **inertia** — resistance to a *change of momentum*,  $F = dp/dt$  (reducing to  $F \approx m \cdot a$  only for  $v \ll c$ ).

So the Position currency splits into two coupled sub-prices — **Space** (the exclusion/extent face) and **Mass** (the context/inertia face). This appendix makes that duality explicit and connects it to the rest of the corpus.

## 2. The rigorous anchor — the Compton wavelength (established physics)

This section is **not** speculative. It is textbook relativistic quantum mechanics, and it is the load-bearing core on which the interpretation rests.

The **Compton wavelength** of a particle of mass  $m$  is:

$$\begin{aligned}\lambda_C &= \hbar / (m \cdot c) && \text{(the REDUCED Compton wavelength)} \\ \lambda_C &= h / (m \cdot c) = 2\pi \cdot \lambda_C && \text{(the ordinary Compton wavelength)}\end{aligned}$$

It is the **characteristic scale at which a fixed-particle-number, one-particle description becomes unreliable** (QFT particle-production becomes relevant) — **not** an unconditional, Lorentz-invariant minimum width of every possible state. Sketch: to pin a particle to a region of size  $\Delta x$  you must inject momenta of order  $\hbar/\Delta x$  (uncertainty principle), i.e. energies of order  $\hbar c/\Delta x$ . Once  $\Delta x \lesssim \lambda_C$ , that energy reaches  $\sim m \cdot c^2$  — enough to **pair-create** — and the very idea of "one particle at a point" dissolves. Consequences (within the one-particle description):

- $m \rightarrow 0 \Rightarrow \lambda_C \rightarrow \infty \Rightarrow$  **no rest-frame one-particle localization scale**. A massless excitation (the photon) has **no rest frame** and no standard sharp one-particle position operator (there is none for massless spin  $> 1/2$ ), so it cannot be pinned as a rest-frame "here." (Nuance, §8: it CAN still be prepared as a finite wave packet and register **local detection events** — masslessness forbids the rest-frame particle-"here", not every local detection.)
- **Larger  $m \Rightarrow$  smaller  $\lambda_C \Rightarrow$  a finer one-particle localization scale**. Mass **sets** that scale (and the inertial response, §3); PMD reads it as the *mass sub-price* of a definite Position.

**Rigorous takeaway (physics, not analogy):** mass **sets the finite one-particle, rest-frame localization scale**  $\lambda_C$  and the inertial response. More mass  $\rightarrow$  a finer rest-frame "here"; zero mass  $\rightarrow$  no rest-frame "here" (local detections still occur). PMD reads this scale as the mass sub-price of Position — the physical fact (the scale) is not in dispute, only the reading is.

**Technical note — reduced scale & localization scope.** Here  $\lambda_C = \hbar/(mc)$  is the **reduced** Compton wavelength (ordinary  $\lambda_C = h/(mc) = 2\pi \cdot \lambda_C$ ). The localization claim is about the breakdown of a **stable one-particle, rest-frame** description: confining a *massive* particle to  $\Delta x \lesssim \lambda_C$  makes particle creation relevant. For *massless* quanta (photons) the point is not that no local detection can occur, but that there is **no rest frame** and **no standard sharp Newton–Wigner one-particle position observable**. PMD reads mass as *enabling a finite rest-frame localization scale*, not as the sole cause of all spatial detection.

### 3. The second face — inertia (the price of *changing* a location)

Newton, relativistically safe:  $F = dp/dt$  with  $p = \gamma \cdot m \cdot v$ , reducing to  $F \approx m \cdot a$  only for  $v \ll c$ . **Inertial mass is resistance to a *change of motion / position*** (acceleration). Placing this beside the canon:

| what is being resisted                                                 | the price (currency)  |
|------------------------------------------------------------------------|-----------------------|
| Form being changed (endurance in time)                                 | <b>Time</b>           |
| <b>occupying</b> a location (exclusion in space)                       | <b>Space</b>          |
| Action being performed (work)                                          | <b>Energy</b>         |
| <b>changing</b> its state of motion (acceleration / $\Delta$ momentum) | <b>Mass (inertia)</b> |

**Note (mass is not a fourth pillar).** This row places **Mass** beside Time/Space/Energy only to display the *parallel structure* of "what is resisted  $\rightarrow$  the price." It does **not** promote mass to a fourth top-level currency. Mass is a **sub-price nested inside the Position pillar** — the inertial/localizability face of Position  $\leftrightarrow$  Space — formalized in §5 as  $P = \sqrt{P_S \cdot P_L}$ , where the mass proxy  $P_M$  is one component of  $P_L$ , not a peer of  $F$ ,  $P$ ,  $A$ . The core triad remains  $U = \sqrt[3]{(F \cdot P \cdot A)}$ . (Proved unique in Part II, §19 / PMD-6, with the elasticity split in §20 / PMD-7 showing the nesting does not double-weight Position.)

So mass plays **two** roles for Position: it sets the *rest-frame localization scale* (§2) **and** the *inertial response to a change of momentum* (§3) — **not** a "cost of moving" (a free body coasts arbitrarily far at constant momentum with no force; mass resists **acceleration**, not displacement). Space answers "*is the system relationally distinguishable / placeable?*"; mass answers "*what is its rest-frame localization scale and inertial response?*" — the spatial face and the inertial face of one currency.

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#### 4. Mass as *concentrated context* (the interpretive leap — L3)

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Why *read* mass as "context"? Because **several** mass contributions and effective inertial parameters are **interaction- or environment-dependent** — *without* this meaning that **all** invariant mass is relational:

- **Mach's principle.** Inertial mass is (on the Machian reading) determined by the distribution of **all other matter** in the universe — inertia is a relation to the cosmic context, not an intrinsic label. (*Status honestly stated: Mach's principle is a real, historically central idea — it motivated Einstein — but it is **not fully established** in GR; frame-dragging realizes it only partially. See §8.*)
- **The Higgs mechanism.** In the Standard Model, elementary fermion and weak-boson masses arise from **coupling to the nonzero Higgs vacuum expectation value** — an *interaction-generated* mass term, **not** friction or drag through a medium. A field without such a coupling gets no Higgs mass term. PMD reads this *interaction origin* of some mass as context — but the Higgs mechanism does not by itself establish that reading, and most of the mass of ordinary matter is **QCD dynamics, not Higgs** (§8).
- **Relational position** (*philosophical, not a physics result*). A "here" is defined only **against** a context of other things; with no context, "here" has no referent.

**Scope (honest).** Mach's principle is an *open* hypothesis (only partially realized in GR via frame-dragging); the Higgs mechanism is *specific and established*; relational position is *philosophical*. PMD interprets the **interaction-dependence** of mass as context — it does **not** claim that *all* invariant mass is Machian or relational.

**U-Theory reading (speculative, L3):** *mass is what a system pays to be **counted as somewhere** by the rest of the universe.* Concentrate context → localizability and inertial response appear together; remove context → the system delocalizes and its "here" dissolves. This is **APPENDIX\_MMT** at the Position axis: MMT says **matter = crystallized meaning**; PMD says, more specifically, **mass = crystallized context** — the Position-currency's share of that crystallization. (*Epigraph "begun to weigh" is figurative; the body claim is inertial mass / rest-frame localizability, not gravitational weight.*)

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## 5. The dual price of Position (the law)

### THE POSITION LOCALIZATION LAW (PMD)

A definite location is bought with two non-substitutable sub-prices of the Position currency:

(i) **Space — constrained spatial support / distinguishability**: become relationally placeable — a constrained spatial support and a context-resolved "where." (*Exclusion / non-overlap is **one** realization, not the universal definition — bosons and fields can share support.*)

(ii) **Mass — localizability / inertia**: in the massive rest-frame sector, set the localization scale (reduced Compton  $\lambda_C = \hbar/mc$  finite) and the **inertial response to a change of momentum** ( $F = dp/dt$ , i.e.  $F \approx m \cdot a$  for  $v \ll c$ ) — *not* a resistance to displacement itself.

**The zero-mass limit is NOT automatically "delocalized" (v1.2.1 correction).** At  $m \rightarrow 0$  the **massive rest-frame localization channel  $P_M$  vanishes** ( $\lambda_C \rightarrow \infty$ ; no rest frame; no rest-frame "here"; **zero invariant rest mass**, yet still carrying energy-momentum and gravitating, at the speed of light). But the overall localizability meter  $P_L$  need **not** be zero — a massless quantum is still scored through the detection channel  $P_D$  (the massless adapter  $g_0$ ), so it is **not** misclassified as zero-Position. A genuinely *unpaid* Position requires **both**  $P_S \rightarrow 0$  and  $P_L \rightarrow 0$ . **Localization is the crystallization of context into a finite localization scale — carried by mass in the massive rest-frame sector, and by the detection channel for massless quanta.**

| face of Position                                               | currency                                     | physics anchor                                     | the zero-limit (unpaid)                                                                                                                                                                  |
|----------------------------------------------------------------|----------------------------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>be relationally placeable</b>                               | Space (spatial support / distinguishability) | constrained support; context-resolved placement    | no distinguishable "where" — the canon's $P \rightarrow 0$                                                                                                                               |
| <b>be localizable &amp; resist <math>\Delta</math>momentum</b> | Mass (context/inertia)                       | $\lambda_C = \hbar/mc$ ; $F = dp/dt$ ; Mach; Higgs | $\lambda_C \rightarrow \infty$ : the <b>mass channel <math>P_M</math> vanishes</b> (no rest-frame "here", no rest inertia; still carries $E - p$ ) — but $P_L$ via $P_D$ need not vanish |

Both faces must be paid for a system to *have a position at all*; a system rich in one and starved of the other has an **ill-defined** Position — which, under  $U = \sqrt[3]{(F \cdot P \cdot A)}$ , drags the whole product down (non-compensation).

**Operational placeholder for "context" (upgraded in Part II).** In this appendix, **context** is a *working label* for the interaction / environment dependence that makes a system's "here" and inertial response well-defined. The minimal placeholder is:

```
Context_proxy := { couplings, boundary conditions, reference frame,
                    and co-present degrees of freedom that enter the
                    system's localization or inertial description }
```

This is **not** a claim that context is a new observable, nor that it equals invariant mass. **Part II (§10, §14) upgrades this placeholder to a concrete definition:** context := the **measurement channel**  $\mathcal{C}$  itself (  $p(y|x, \mathcal{C})$  ); its *localization content* is quantified by the **Fisher information**  $J_{\mathcal{C}}$  about the position parameter  $x$ , with additivity across independent channels (PMD-3) and a data-processing bound (PMD-4). (*The channel is the object;  $J_{\mathcal{C}}$  is how much it tells you about **where**.*) Every numerical use of PMD must still declare its own proxy explicitly (§8).

**Nesting (keeps Position as ONE pillar — not a fourth price), with a domain gate.** The two faces compose *inside* Position, geometrically, so a zero in either zeros Position. The localizability face is a **general localization/detection meter**  $P_L$ , defined (**v1.4, adopting §17**) as a weighted geometric blend of a **detection channel**  $P_D$  and the **mass-capacity proxy**  $P_M$ , with the **pre-registered default**  $\eta = \frac{1}{2}$ :

```
POSITION ↔ SPACE
  | P_S : constrained spatial support / relational distinguishability      ∈ [0,1)
  | P_L : localizability / localization-detection meter                  ∈ [0,1)
    | massive rest-frame sector : P_L = P_D^(1-η) · P_M^η ,   η = ½
default  (§17)
    | P_D = detection/Fisher channel (§15) ; P_M = g_M(L*/λ_C) (§16)
    | special case η = 1 → P_L = P_M    (mass-only, legacy Compton reading)
    | massless field sector      : P_L = P_D = g_0( Pr[detect in R,Δt], σ_R, frame )

P = P_PMD = √( P_S · P_L )          U = √[3]( F · P_PMD · A )
```

This blend discharges the two key negative controls **by construction** (§8, §17): a *massive-but-delocalized* state has  $P_D = 0 \Rightarrow P_L = 0$ ; a *massless-but-detected* quantum has  $P_L = P_D > 0$  (never zero-Position). Mass is **not** a fourth top-level pillar; it is **one component** (via  $P_M$ ) of the nested  $P_L$  meter, and the mass-only  $P_L = P_M$  reading survives as the  $\eta = 1$  special case.  $g_M$ ,  $g_0$ ,  $\eta$ , the reference scale  $L^*$ , the normalization, and a missing-data rule are **declared before use** and are **calibration candidates, not canon** (§8). (*Part II §16–§17 derives the admissible forms and the adapter;  $\eta$  must be pre-registered, never fit post hoc.*)



## 6. The conjugacy — position, momentum, and the massless limit

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A **massless** excitation (e.g. a photon) has **no rest frame** and **no massive-particle rest-frame localization scale** — it is not a little ball with a coordinate. But it is **not** "everywhere": a single-photon **wave packet** can be prepared with finite spatial-temporal localization and produce **local detection events** (there is simply no sharp Newton–Wigner one-particle position observable for it). So masslessness removes the *rest-frame particle-"here"*, **not all localization**. Acquiring mass gives a massive particle a rest-frame localization scale ( $\lambda_C$ ) and an inertial response — it does **not** follow that massless quanta cannot be localized at all.

This mirrors the **position-momentum uncertainty**  $\Delta x \cdot \Delta p \geq \hbar/2$ : a sharp "here" ( $\Delta x$  small) costs a large momentum spread. **The conjugate of position is momentum, not mass** — but mass enters the trade through the relativistic dispersion  $E^2 = p^2 c^2 + m^2 c^4$  and the localization scale  $\lambda_C = \hbar/mc$ . (In general  $p = \gamma \cdot m \cdot v$ ; a photon carries  $p = E/c$  despite **zero** rest mass — so  $p = m \cdot v$  is only the  $v \ll c$  approximation and is never applied here to massless or relativistic cases.) Sharpening a "here" costs momentum spread; for a **massive** particle **mass sets the rest-frame scale** of that trade. But it is **momentum, not mass**, that is conjugate to position — and a massless quantum can still be wave-packet-localized. The defensible statement is **not** "delocalized  $\leftrightarrow$  massless"; it is only that mass **sets a massive particle's rest-frame localization scale**.

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## 7. Binding it to the whole corpus

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- **SSS** /  $U = \sqrt[3]{(F \cdot P \cdot A)}$ : the **Position pillar** now carries an explicit *dual meter* — spatial support / distinguishability **and** localizability (with mass as one proxy). A system with strong Form and Action but near-zero relational placement / localizability has a near-zero Position, so  $U$  collapses. This is the **physical face** of the corpus's central thesis (autobiography, POS): *strong Form and Action do not compensate a Position near zero. Illustrative flavour only (not a classification):* a system rich in Form and Action but poor in relational placement / localizability scores low on Position and, by non-compensation, low on  $U$ . **This must not be read literally for a photon** — a massless quantum is locally detectable and is *not* zero-Position (§5 adapter  $g_0$ , §8).
- **MMT**: matter = crystallized meaning; PMD refines the Position axis — **mass = crystallized context** (*interpretive*). The Big Bang as maximal concentration ( $\max \mathcal{M}$ ) is, *under the Machian reading* (§4, §8 — an open question, not settled physics), context concentrating until inertial response appears; without that reading, the same event is simply maximal energy density, and PMD adds no extra claim.
- **DIM**: required meaning scales with dimension; localization is a *spatial-dimensional* act, and PMD says the massive rest-frame sector of that act is paid partly in mass/context.
- **QMC (measurement)**: a **localized position measurement records a localized outcome** and, under standard state-update modelling, **conditionally prepares a more localized post-measurement state** — via an interaction (context) with an apparatus. PMD reads that interaction as paying the context price, without committing to any one interpretation of measurement. (*Formalized in Part II §11–§15: the apparatus channel contributes Fisher information  $J_C$  about position.*)



- **POS (Principle of Sequence):**  $F \rightarrow P \rightarrow A$ . Physically, a system needs **Form** (persistence in time) and then a **Position** (a "here," bought with space **and**, in the massive sector, mass/context) *before* it can spend **Energy** in Action. No "here"  $\rightarrow$  no base to act from — the Forbidden Action at the level of physics.

## 8. Falsifiability and honest scope

**What is rigorous (not speculative):**

- $\lambda_C = \hbar/mc$  (reduced) as the **one-particle, rest-frame** localization scale;  $m \rightarrow 0 \Rightarrow$  no rest-frame localization scale,  $m \uparrow \Rightarrow$  a finer one — standard relativistic QM.
- Inertia as resistance to a *change of momentum* ( $F = dp/dt$ ;  $F \approx m \cdot a$  only for  $v \ll c$ ) — standard mechanics.
- Mass from Higgs coupling — Standard Model. **Caveat:** the Higgs gives **elementary** rest masses; **most** of the mass of ordinary matter (protons, neutrons) is **QCD binding / field energy**, not a Yukawa coupling. PMD reads both as *context becoming inertial* — but only the Higgs part is a *direct* SM mass-generation mechanism. *Why the composite (QCD) case strengthens the reading rather than weakening it:* nucleon mass is not carried by its constituents in isolation — it is the **energy of the internal interactions** (gluon fields, confined quark motion) of the system *with itself*. A proton's inertia is therefore the inertial face of its own **internal relational context** — the "mass = crystallized context" reading applied to a system's interactions with its own parts, exactly the composite it is meant to describe. The Higgs case is *interaction with an external field* (the VEV); the QCD case is *interaction of the system with itself* — both are context becoming inertial, and the second is the more direct instance. (*Avoid "weight" language here: the claim is about inertial mass / rest energy, not gravitational weight. The signed mass-defect subtlety — QCD adds mass, nuclear/atomic binding subtracts it — is formalized in Part II §22 / PMD-8 as  $\chi_{int}$ .*)

**What is speculative interpretation (L3/L4), and its honest hooks:**

- **"Mass = concentrated context / the Position-currency's inertia face"** is a *reading*, not a derivation. It does not add to or modify GR/QFT; it re-describes them in F/P/A terms.
- **The strong converse — "no location  $\Rightarrow$  no mass" — is NOT standard physics and is flagged as the most speculative claim.** A *delocalized but massive* state exists in ordinary QM (a momentum eigenstate of a massive particle is spread over all space yet has mass). PMD's honest claim is the *rigorous direction only*: **masslessness forbids the rest-frame particle-"here"** (no rest frame; no sharp one-particle position operator for the photon), **while mass enables a finite localization scale**; local detection of massless quanta still occurs. The reverse (delocalized  $\Rightarrow$  massless) is offered as a suggestive lens, not asserted.
- **Mach's principle is genuinely open.** If inertia were shown to be entirely independent of the cosmic matter distribution, the "context confers mass" reading weakens to a metaphor. If frame-dragging / relational-inertia results strengthen the Machian picture, it strengthens. PMD is therefore **hostage to a real open question in physics**, and says so.

**Null model, empirical content, and required controls (RH).**

- **Present empirical content (honest):** as of v1.5, PMD makes **no observation that standard  $\lambda_C$ , relativistic kinematics, QFT, and GR do not already make** — the one concrete falsifiable ceiling it now states ( $\text{Var}(\hat{x}) \geq \hbar^2/(8NmK)$ , §12.1) is a consequence of *standard* quantum estimation theory, not new physics. The functions  $g_M$ ,  $g_0$ , and the scale  $L^*$  are calibration candidates, not fitted laws. This is not a soft admission — Part II **proves** it: the **no-go theorem (§23 / PMD-9)** shows that if the PMD context variable is any deterministic function of the standard variables,  $C = f(X_{\text{std}})$ , then  $I(Y; C | X_{\text{std}}) = 0$  for every observable  $Y$ . **PMD earns empirical content only against an independently measured context observable  $C_R$  with  $I(Y; C_R | X_{\text{std}}) > 0$ .**
- **Null model:**  $H_0$ : every observation is explained by the standard  $\lambda_C$ ,  $p^u$ , and QFT/GR, with no PMD context variable.
- **What would count as content:** a mass/localization meter that is (i) independently measurable, (ii) adds predictive power over  $H_0$ , and (iii) never misclassifies massless, locally-detected systems.
- **Required negative controls before any use:** (a) **massive-but-delocalized** (a momentum eigenstate — massive yet spread out → refutes "delocalized  $\Rightarrow$  massless"); (b) **massless-but-locally-detected** (a photon wave packet → refutes  $P_L = 0$  for massless); (c) **inertial-motion null** (constant-velocity coasting, no force → refutes "mass resists displacement"); (d)  **$P_S$ -only rival** (must the mass sub-meter beat a Position score built from spatial support alone?). *(These are discharged by construction in the upgraded adapter, Part II §17.)*

**Epistemic level (Part I): L3/L4 speculative** — a coherent, corpus-consistent *interpretation* resting on a rigorous physical core. It makes no claim of empirical proof and adds no new governing equation. It sharpens the meaning of the Position currency; it does not re-derive physics. *(The formal core that IS provable — the localization–energy bound, Fisher additivity, the nested-mean uniqueness, and the no-go — is Part II.)*

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## PART II —

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## FORMAL LAYER (PMD-MATH)

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**From philosophical analogy to a formal measurement framework** — what can be *proved*, what is a *modelling choice*, and what is *forbidden* without a new observable.

**One-line thesis (the strongest defensible claim).** In the massive non-relativistic sector, **invariant mass upper-bounds the spatial Fisher information extractable per unit kinetic energy** ( $F_Q/K \leq 8m/\hbar^2$ ); the **measurement context** determines what fraction of that capacity is actually realized ( $J_C \leq F_Q$ ). Therefore **mass sets capacity; context realizes accessibility** — two distinct, complementary roles, *not* an identity.

## 9. Consistency audit

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### 9.1 What CAN be proved mathematically (Part II)

- the reciprocal identity  $m \cdot \lambda_C = \hbar/c$  (§11 / PMD-1);
- the mass–energy–localization inequality  $F_Q \leq 8mK/\hbar^2$  (§12 / PMD-2);
- additivity of context Fisher information across independent channels (§13 / PMD-3);
- the data-processing (monotonicity) bound: losing context cannot raise position information (§14 / PMD-4);
- the admissible one-parameter family for  $g_M$  from a log-odds axiom (§16 / PMD-5);
- uniqueness of the nested geometric mean under separability + symmetry + idempotence (§19 / PMD-6);
- the elasticity split showing the nesting does **not** double-weight Position (§20 / PMD-7);
- the exact participation of interactions in composite invariant mass (§22 / PMD-8);
- the condition for empirical surplus over the null model, as a **no-go** (§23 / PMD-9);
- the **exact relativistic** localization bound, of which the NR bound is the leading term (§12.1 / PMD-2R);
- the **capacity–efficiency factorization**  $J_C = \epsilon \cdot u \cdot (8mK/\hbar^2)$  (§15.1 / PMD-10);
- **per-meter reference-scale invariance** — and its **failure for the combined score** (§16.1 / PMD-11);
- the **surplus identity**  $\mathbb{E}[\Delta \log\text{-loss}] = I(Y; C_R | X_{\text{std}})$  (§23.1 / PMD-12).

### 9.2 What CANNOT be proved by mathematics alone

The following stay **interpretive (L3/L4)** and are *not* elevated by Part II:

```
mass = context
localization = crystallization of context
cosmic context causes all inertia
```

Mathematics can derive consequences *once a formal definition of "context" is accepted*, but it cannot prove that definition corresponds to a fundamental physical reality.

### 9.3 Residual wording defect in Part I (fixed in v1.2.1)

Part I §5's law box once carried "the unpaid limit is the massless, **delocalized** state," while §6/§8 correctly reject "massless  $\Leftrightarrow$  delocalized." The mathematically correct statement — now enforced in §5 — is: at  $m \rightarrow 0$  the **massive rest-frame channel**  $P_M$  **vanishes**, but the overall localizability meter  $P_L$  need **not** be zero (it is carried by the detection/Fisher channel  $P_D$ ). Masslessness removes the rest-frame particle-"here", not all localization.

---

## 10. Formal domain

Let:

$L^* > 0$  a pre-registered, task-specific localization scale [m]  
 $m \geq 0$  invariant mass [kg]  
 $\lambda_C = \hbar/(mc)$  reduced Compton wavelength (for  $m > 0$ ) [m]  
 $x \in \mathbb{R}^d$  spatial-translation parameter (the "where")  
 $\rho_x$  quantum state after translation by  $x$   
 $\mathcal{C}$  the measurement context  
 $P_S, P_D, P_M, P_L, P \in [0, 1]$  ( $P_D, P_S, P_M$  lie strictly in  $[0, 1]$  – they saturate toward 1, never reach it)

Spatial translation is generated by momentum  $\hat{p}$ :

$$\rho_x = e^{(-i x \cdot \hat{p} / \hbar)} \cdot \rho_0 \cdot e^{(+i x \cdot \hat{p} / \hbar)}$$

The context  $\mathcal{C}$  is represented by a POVM / classical measurement channel:

$$p(y \mid x, \mathcal{C}) = \text{Tr}(\rho_x \cdot E_y^{\mathcal{C}})$$

so **"context" stops being a word**. It is:

$$\mathcal{C} = \{ \text{coupling, boundary, frame, measurement channel} \}$$

— and it fixes *how much information about  $x$  can actually be extracted*.

## 11. Theorem PMD-1 — Compton reciprocity · [L1: identity]

For  $m > 0$ :  $\lambda_C = \hbar/(mc)$ , hence

$$\blacksquare \quad m \cdot \lambda_C = \hbar / c \quad (\text{mass} \times \text{localization scale is a universal constant})$$

**Proof.**  $m \cdot \lambda_C = m \cdot \hbar/(mc) = \hbar/c$ .  $\blacksquare$  Also  $d\lambda_C/dm = -\hbar/(cm^2) < 0$ , so the Compton scale is strictly decreasing in mass.

**Non-relativistic inertial corollary.** At fixed force  $F$  and  $v \ll c$ ,  $a = F/m$ , so

$$\lambda_C / a = (\hbar/mc) / (F/m) = \hbar / (cF) \quad \Rightarrow \quad \blacksquare \quad \lambda_C \propto a \propto 1/m \quad (\text{at fixed } F)$$

**Reading (not a proof of "mass = context").** The *same* parameter  $m$  that shrinks the massive rest-frame localization scale also shrinks the acceleration response at fixed force. The two PMD anchors — localization scale (§2) and inertia (§3) — are governed by **one** physical quantity. That is a genuine structural unification; it is *not* a claim that mass is context.

## 12. Theorem PMD-2 — Localization Information–Energy bound · [L2: proved, non-relativistic]

(The strongest new formal result — the defensible quantitative core of PMD.)

For the translation family  $\rho_x$  generated by  $\hat{p}$ , the **quantum Fisher information** for estimating  $x$  obeys

$$F_Q(\rho_x) \leq 4 (\Delta p)^2 / \hbar^2 \quad (\text{equality for pure states})$$

With non-relativistic kinetic energy  $K = \langle p^2 \rangle / (2m)$  and  $(\Delta p)^2 = \langle p^2 \rangle - \langle p \rangle^2 \leq \langle p^2 \rangle = 2mK$ :

$$\blacksquare \quad F_Q \leq 8 m K / \hbar^2 \quad (\text{equality for a pure, zero-mean-momentum state})$$

**Proof.** Combine  $F_Q \leq 4(\Delta p)^2/\hbar^2$  (QFI of a unitary family = 4·Var of its generator, with  $\leq$  for mixed states) and  $(\Delta p)^2 \leq 2mK$ . ■

**Cramér–Rao corollary.** For  $N$  independent measurements,  $\text{Var}(\hat{x}) \geq 1/(N \cdot F_Q)$ , hence

$$\blacksquare \quad \text{Var}(\hat{x}) \geq \hbar^2 / (8 N m K)$$

**Capacity reading (defensible).** Per unit kinetic-energy budget, larger mass permits a lower fundamental error floor on estimating a spatial displacement:

$$\blacksquare \quad F_Q / K \leq 8 m / \hbar^2 \quad (\text{"mass = localization-information leverage per unit energy"})$$

This is far more defensible than "localization is bought with mass": it is a *proved inequality*, explicitly scoped to the non-relativistic massive sector, that assigns mass a precise operational role (capacity), not a metaphysical one. **Scope:** as stated the bound is **non-relativistic** (it uses  $K = \langle p^2 \rangle / 2m$ ). §12.1 now supplies the **exact relativistic form**, of which this is the leading term.

### 12.1. Theorem PMD-2R — exact relativistic localization bound · [L2: proved, exact]

(New in v1.5. Closes the non-relativistic scope gap of PMD-2 — the bound is not merely an NR approximation but the leading term of an exact result.)

Spatial translation is generated by  $\hat{p}$  **at all velocities**, so  $F_Q \leq 4 \cdot \text{Var}(\hat{p}) / \hbar^2 \leq 4 \langle \hat{p}^2 \rangle / \hbar^2$  holds relativistically as well. For a free particle the dispersion is an **operator identity**,  $\hat{H}^2 = \hat{p}^2 c^2 + m^2 c^4$ , hence

$$\langle \hat{p}^2 \rangle = ( \langle \hat{H}^2 \rangle - m^2 c^4 ) / c^2 \quad (\text{exact} - \text{no expansion})$$

and therefore

$$\blacksquare \quad F_Q \leq 4( \langle \hat{H}^2 \rangle - m^2 c^4 ) / (\hbar^2 c^2) \quad (\text{exact, all velocities})$$

**Proof.**  $\text{Var}(\hat{p}) \leq \langle \hat{p}^2 \rangle$ ; substitute the operator identity.  $\blacksquare$

### Scope — four hypotheses, all load-bearing

1. **Single particle.**  $\hat{H}^2 = c^2 \hat{p}^2 + m^2 c^4$  is an operator identity **only** on a one-particle positive-energy Poincaré irrep. For  $N > 1$  it is **false**: the correct object is the *invariant-mass operator*  $\hat{M}^2 c^4 \equiv \hat{H}^2 - c^2 \hat{p}^2$ , which acts nontrivially on relative momenta. *Counterexample*: two free particles of mass  $m$  with momenta  $\pm p$  have  $\hat{p}|\psi\rangle = 0$ , so  $\langle \hat{p}^2 \rangle = 0$ , yet  $(\langle \hat{H}^2 \rangle - M^2 c^4)/c^2 = 4p^2 \neq 0$  — the formula overstates  $\langle \hat{p}^2 \rangle$  without bound. It also fails for interacting  $\hat{H} = \hat{H}_{\text{free}} + \hat{V}$  and for general Fock states.
2. **Positive-energy sector.** On the full Dirac spectrum a negative-energy state has  $\langle E_{\text{kin}} \rangle < 0$  and the expansion below loses its sign.
3. **Finite second moment,**  $\langle \hat{H}^2 \rangle < \infty$ ; otherwise the bound is true but vacuous.
4. **"Exact" qualifies the identity, not the bound.** The inequality discards  $\langle \hat{p} \rangle^2$ , so it is loose by exactly  $4\langle \hat{p} \rangle^2 / \hbar^2$  — a **frame-dependent** slack that grows like  $\gamma^2$  under boosts. It is saturable only in the zero-mean-momentum frame.

**Relation to PMD-2 — and a correction to the naive reading.** Writing  $\hat{E}_{\text{kin}} \equiv \hat{H} - mc^2$  and  $(\Delta H)^2 = \langle \hat{H}^2 \rangle - \langle \hat{H} \rangle^2$ , the algebra is exact (on the positive-energy sector, where every term is non-negative):

$$\langle \hat{H}^2 \rangle - m^2 c^4 = 2mc^2 \langle E_{\text{kin}} \rangle + \langle E_{\text{kin}} \rangle^2 + (\Delta H)^2$$

$$\blacksquare \quad F_Q \leq \underbrace{8m\langle E_{\text{kin}} \rangle / \hbar^2}_{\text{PMD-2 form}} + \underbrace{4( \langle E_{\text{kin}} \rangle^2 + (\Delta H)^2 ) / (\hbar^2 c^2)}_{O(1/c^2), \geq 0}$$

### ⚠ This does **NOT** make $8mK/\hbar^2$ a relativistic ceiling — it proves the opposite

The extra terms are **non-negative and added to the right-hand side**, so the relativistic ceiling **exceeds** the PMD-2 expression. A larger ceiling does not validate the smaller one; it *removes* its status as a bound.  **$F_Q \leq 8mK/\hbar^2$  is therefore NOT valid relativistically when  $K$  denotes relativistic kinetic energy — it is violated at every nonzero momentum.**

**Exact violation factor.** For the zero-mean state  $(|+p\rangle + |-p\rangle)/\sqrt{2}$  one has  $F_Q = 4p^2/\hbar^2$  exactly, while  $E_{\text{kin}} = mc^2(\sqrt{1+u^2} - 1)$  with  $u \equiv p/(mc)$ . Then

$$\blacksquare \quad F_Q / (8m \cdot E_{\text{kin}}/\hbar^2) = u^2 / [2(\sqrt{1+u^2} - 1)] = (\sqrt{1+u^2} + 1) / 2 > 1 \quad \text{for every } u > 0$$

| $u = p/mc$       | 0.01     | 0.1    | 1            | 10          | 100          |
|------------------|----------|--------|--------------|-------------|--------------|
| violation factor | 1.000025 | 1.0025 | <b>1.207</b> | <b>5.52</b> | <b>50.50</b> |

(Closed form verified against direct computation; as  $m \rightarrow 0$  the PMD-2 expression  $\rightarrow 0$  while  $F_Q$  stays finite — violation by an unbounded factor.)

**Corrected status of PMD-2.**  $F_Q \leq 8mK/\hbar^2$  holds **if and only if**  $K$  is defined as  $\langle \hat{p}^2 \rangle / 2m$  — in which case it is not an independent physical bound at all but a *rewriting* of  $F_Q \leq 4\langle \hat{p}^2 \rangle / \hbar^2$  using the NR definition of kinetic energy. Its content is the **identification** of  $\langle \hat{p}^2 \rangle$  with a mass-times-energy budget, and that identification is exactly what fails relativistically. PMD-2 is thus **strictly non-relativistic**, and PMD-2R is the correct statement whenever velocities are not small.

*(This correction was forced by adversarial review of the v1.5 draft, which had read the expansion backwards; the error and its repair are recorded here rather than silently removed.)*

### ⚠ Honest consequence — the "capacity" reading is budget-dependent

The direction of the mass effect depends on **which energy budget is held fixed** — and the contrast is *not* relativistic-vs-non-relativistic:

fixed KINETIC budget  $K = \langle \hat{p}^2 \rangle / 2m$  : ceiling =  $8mK/\hbar^2$   
 $\rightarrow$  INCREASES with  $m$   
fixed KINETIC budget  $K = \langle \hat{H} \rangle - mc^2$  : ceiling =  $4(2mc^2K + K^2 + (\Delta H)^2)/(\hbar^2c^2) \rightarrow$  INCREASES with  $m$   
fixed TOTAL budget  $\langle \hat{H}^2 \rangle$  : ceiling =  $4(\langle \hat{H}^2 \rangle - m^2c^4)/(\hbar^2c^2)$   
 $\rightarrow$  DECREASES with  $m$

So the reversal is driven entirely by **whether rest energy sits inside the budget**, not by relativistic kinematics: at fixed *kinetic* energy the direction is the same in both regimes. **"More mass = more localization capacity" holds under a kinetic-energy budget and reverses under a total-energy budget**, because rest energy consumes what would otherwise buy momentum spread.

**Necessary technical care.** The total-energy budget must be stated as fixed **second moment**  $\langle \hat{H}^2 \rangle$ , *not* fixed mean  $\langle \hat{H} \rangle$ . Since  $\langle \hat{H}^2 \rangle = \langle \hat{H} \rangle^2 + (\Delta H)^2$  and  $(\Delta H)^2$  is unbounded at fixed mean, **the ceiling is unbounded above at fixed  $\langle \hat{H} \rangle$**  (explicitly: a two-component superposition holding  $\langle \hat{H} \rangle = 2mc^2$  while pushing the upper branch up gives  $\langle \hat{H}^2 \rangle - m^2c^4 = 4, 11, 101, 10^4, 10^8 \dots$  in units  $m = c = 1$ ). Writing the budget as  $E^2 - m^2c^4$  with  $E = \langle \hat{H} \rangle$  — as the v1.5 draft did — is an error.



Any application of PMD **must declare which budget is held fixed**; the §30 Localization-Leverage statement is scoped to the fixed-kinetic-energy reading.

**One concrete falsifiable ceiling (answers "PMD makes no checkable statement").**

Combining PMD-2R with the Braunstein–Caves inequality ( $J_C \leq F_Q$  for any POVM) gives a hard design constraint on **every** localization protocol — any measurement, any number of detectors, any classical post-processing:

$$\blacksquare \quad J_C \leq 4\langle \hat{p}^2 \rangle / \hbar^2 \leq 4(\langle \hat{H}^2 \rangle - m^2 c^4) / (\hbar^2 c^2) \quad \Rightarrow \quad \text{Var}(\hat{x}) \geq \hbar^2 c^2 / (4N(\langle \hat{H}^2 \rangle - m^2 c^4))$$

(non-relativistically, with  $K \equiv \langle \hat{p}^2 \rangle / 2m$ :  $\text{Var}(\hat{x}) \geq \hbar^2 / (8NmK)$ )

(one-particle, free, positive-energy,  $\langle \hat{H}^2 \rangle < \infty$ ; the  $\langle \hat{p}^2 \rangle$  form is the primitive one and is what a metrology test actually constrains)

This **is** experimentally falsifiable: a certified localization variance below the floor, at a certified energy budget, refutes it. **Honest label:** it follows from *standard* quantum estimation theory — it is **not new physics**. PMD's contribution is to *identify this ceiling as the Position-axis capacity* and to build the meter around it. PMD-9 (§23) still blocks any claim of *novel* predictive content.

### 13. Theorem PMD-3 — context Fisher information is additive · [L2: proved]

For  $n$  conditionally independent channels,  $p(y_1, \dots, y_n \mid x) = \prod_k p_k(y_k \mid x)$ , with per-channel classical Fisher information  $J_k(x) = \mathbb{E}[(\partial_x \ln p_k)^2]$ :

$$\blacksquare \quad J_C(x) = \sum_k J_k(x)$$

**Proof.** The score of the product is the sum of scores,  $s = \sum_k s_k$ . Then  $J_C = \mathbb{E}[(\sum s_k)^2] = \sum_k \mathbb{E}[s_k^2] + 2 \sum_{\{i < j\}} \mathbb{E}[s_i s_j]$ . For regular models  $\mathbb{E}[s_k] = 0$ ; under conditional independence  $\mathbb{E}[s_i s_j] = \mathbb{E}[s_i] \mathbb{E}[s_j] = 0$ . Hence  $J_C = \sum_k J_k$ . ■

**Interpretation.** This gives a precise meaning to "more context": *more independent relational channels  $\Rightarrow$  more position information*. It does **not** say context creates mass — only that relational context increases the **measurability** of "where."

### 14. Theorem PMD-4 — losing context cannot raise localization information · [L2: proved]

For a coarse-graining Markov chain  $x \rightarrow Y \rightarrow Z$ :

$$\blacksquare \quad J_Z(x) \leq J_Y(x) \quad (\text{Fisher-information data-processing inequality})$$

**Proof (sketch, under standard regularity** — support of  $Y$  independent of  $x$ , and  $\partial/\partial x$  interchangeable with integration<sup>\*\*</sup>).<sup>\*\*</sup> The coarse-grained score is a conditional expectation,  $s_Z = \mathbb{E}[s_Y \mid Z]$ ; by Jensen,  $\mathbb{E}[s_Z^2] = \mathbb{E}[\mathbb{E}[s_Y^2 \mid Z]] \leq \mathbb{E}[s_Y^2]$ , i.e.  $J_Z \leq J_Y$ . ■

**The defensible context thesis.** Removing or coarse-graining relational channels **cannot increase** the information with which a system is localizable. (The converse — that adding context *creates* location rather than reveals it — is *not* claimed.)

## 15. Operational localization (detection) meter $P_D$ · [L3: modelling choice]

Define the dimensionless Fisher ratio  $z_D = L^2 \cdot J_C$  and

$$\blacksquare \quad P_D = z_D / (1 + z_D) = L^2 \cdot J_C / (1 + L^2 \cdot J_C)$$

Properties (all verified):  $0 \leq P_D < 1$ ;  $P_D(0) = 0$ ;  $P_D \rightarrow 1$  as  $J \rightarrow \infty$ ;  $\partial P_D / \partial J = L^2 / (1 + L^2 J)^2 > 0$ . It is dimensionless, bounded, strictly monotone, saturating, and **valid for both massive and massless local detections** — a cleaner realization of Part I's massless adapter  $g_0$ .

### 15.1. PMD-10 — the capacity–efficiency decomposition · [L3: a definition + two scoped range claims — NOT a theorem]

(New in v1.5; **demoted in v1.5.1** after review. The decomposition itself is true **by construction**; all of its content sits in the two range claims and their scope.)

Two inequalities chain together — the first is Braunstein–Caves (the classical Fisher information of **any** POVM cannot exceed the quantum Fisher information), the second is PMD-2:

$$J_C \leq F_Q \leq 8mK/\hbar^2 \quad (\text{non-relativistic; use PMD-2R for the exact form})$$

Define two dimensionless ratios, each in  $[0, 1]$  by exactly those two inequalities:

|                           |              |                    |                                                      |
|---------------------------|--------------|--------------------|------------------------------------------------------|
| $\epsilon = J_C / F_Q$    | $\in [0, 1]$ | CONTEXT EFFICIENCY | – how much of what is knowable the channel extracts  |
| $u = F_Q / (8mK/\hbar^2)$ | $\in [0, 1]$ | STATE UTILIZATION  | – how much of the mass-energy ceiling the state uses |

Then, identically:

$$J_C = \varepsilon \cdot u \cdot (8mK/\hbar^2)$$

└ context ┘ └ state ┘ └ mass-energy capacity ┘

**Honesty note (v1.5.1).** As an equation this is  $J_C = J_C$  — a tautology, since  $\varepsilon$  and  $u$  are *defined* as the two ratios. It is a bookkeeping decomposition, not a theorem, and it is **undefined when  $F_Q = 0$**  (e.g. any momentum eigenstate, where  $\text{Var}(\hat{p}) = 0$  at arbitrarily large  $m$  and  $K$ ). All the substantive content is in the two range claims, which must therefore carry their own scope:

- $\varepsilon \in [0,1]$  — **sound**, by Braunstein–Caves, *provided*  $J_C$  is the classical Fisher information of the **same** family  $p_x$ , about the **same** parameter, per **identical resource unit** (a per- $N$ -copy or per-unit-time  $J_C$  compared against a single-shot  $F_Q$  can exceed 1; so can a finite-sample plug-in estimate, which is upward-biased).
- $u \in [0,1]$  — **sound only in the non-relativistic sector with  $K \equiv \langle \hat{p}^2 \rangle / 2m$** . It **fails** for relativistic kinetic energy at every nonzero momentum (§12.1), and for any phenomenological "budget" that is not  $\langle \hat{p}^2 \rangle / 2m$ . *(It does survive one useful generalization: for NR multi-particle centre-of-mass translation, Cauchy–Schwarz gives  $\langle \hat{p}^2 \rangle \leq 2MT$  with  $M = \sum m_i$ , so  $u \leq 1$  holds for the collective generator.)*

**Reading.** Within that scope the three factors name the three genuinely distinct things PMD talks about: *what the world permits* (mass-energy), *what the state offers* (utilization), *what the context extracts* (efficiency). This is the informal content of §30's "mass sets capacity; context realizes accessibility."

### Consequence for the §17 adapter — $\eta$ is not the weight on mass

(The v1.5 draft asserted here that  $P_D^{(1-\eta)} \cdot P_M^\eta$  "double-counts mass" as a proved structural fact. **That claim is withdrawn in v1.5.1** — it does not follow. The bound  $J_C \leq 8mK/\hbar^2$  is one-directional: it constrains only the ceiling, and since  $\varepsilon, u$  range freely over  $[0,1]$ , the realized  $J_C$  is **not a monotone function of  $m$**  — for any mass there are states with  $J_C = 0$ . A ceiling that grows with  $m$  therefore implies nothing about whether the realized  $P_D$  grows with  $m$ . Statistical dependence between  $P_D$  and  $P_M$  is an **empirical** property of the population being scored, not a derivable one; the bound induces only a triangular support region.)

**What does survive, and is checkable.** Because  $P_D$ 's *achievable range* depends on  $m$ , the elasticity of the composite with respect to  $\log m$  is not  $\eta$  but

$$\blacksquare \quad \eta_{\text{eff}} = \eta + (1 - \eta) \cdot s, \quad s \equiv \partial \log P_D / \partial \log m \quad \text{over the scored population, } s \in [0,1]$$

So  $\eta$  may not be read as "the weight on mass": the realized mass weight lies anywhere in  $[\eta, 1]$  and must be **estimated on the corpus** (by measuring  $s$ , or equivalently  $\text{Corr}(\text{logit } P_D, \text{logit } P_M)$ ), never asserted. Pre-registering  $\eta$  without reporting  $s$  leaves the effective mass weight undetermined.

**Calibration rival (unchanged in status).** Blending the two factors that are *not* nested in one another remains a reasonable alternative to test:

$$P_L = P_{\text{cap}}^{(1-\eta)} \cdot P_{\text{eff}}^{\eta}, \quad P_{\text{cap}} = g(L^2 \cdot 8mK/\hbar^2), \quad P_{\text{eff}} = \epsilon = J_C/F_Q$$

**Status:** L3 modelling choice for the §29 model comparison — motivated by interpretability of  $\eta$ , **not** by a proved redundancy.

## 16. Theorem PMD-5 — the admissible family for $g_M$ · [L3: conditional derivation]

Let  $x_M = L^*/\lambda_C = mcL^*/\hbar = m/m^*$ , with reference mass  $m^* = \hbar/(cL^*)$ . Require  $g_M : (0, \infty) \rightarrow (0, 1)$  continuous, strictly increasing,  $g_M(1) = 1/2$ , and **log-odds additive under multiplicative rescaling** of  $x_M$ :

$$\ell(x \cdot y) = \ell(x) + \ell(y), \quad \text{where } \ell(x) = \ln[ g_M(x) / (1 - g_M(x)) ]$$

Then the continuous solutions of that Cauchy equation are  $\ell(x) = \alpha \cdot \ln x$  ( $\alpha > 0$ ), so

$$\blacksquare \quad g_M(x) = x^\alpha / (1 + x^\alpha)$$

**Proof.**  $\ell(xy) = \ell(x) + \ell(y)$  continuous  $\Rightarrow \ell(x) = \alpha \ln x$ ; exponentiating  $g/(1-g) = x^\alpha$  and solving gives the logistic form. ■

**Preferred Fisher exponent.** If the mass channel is read as an inverse-variance capacity  $J_M \propto 1/\lambda_C^2$ , the natural exponent is  $\alpha = 2$ , giving

$$\blacksquare \quad P_M = (L^*/\lambda_C)^2 / (1 + (L^*/\lambda_C)^2) = (m/m^*)^2 / (1 + (m/m^*)^2)$$

This is **not** a new physical law — it is a canonical normalization candidate justified by dimensional analysis, inverse-variance reading, boundedness, and scale symmetry.  $\alpha \in \{1, 2, \text{free}\}$  is a calibration/model-comparison question, not settled.

## 16.1. Theorem PMD-11 — reference-scale invariance: what survives and what does not · [L2: proved (both directions)]

(New in v1.5. The reference scale  $L^*$  is a declared convention, so it is essential to know exactly which PMD conclusions depend on it. One half of this result is positive; the other half is a **negative result** that constrains how PMD may be used.)

### (a) Per-meter — invariance holds (positive)

Under a change of reference scale  $L^* \rightarrow \lambda L^*$  ( $\lambda > 0$ ), each meter's log-odds shifts by a **constant that is the same for every system**:

$$\begin{aligned}\text{logit } P_M(m; \lambda L^*) &= \text{logit } P_M(m; L^*) + \alpha \cdot \ln \lambda \\ \text{logit } P_D(J; \lambda L^*) &= \text{logit } P_D(J; L^*) + 2 \cdot \ln \lambda\end{aligned}$$

Therefore, for **any two systems** compared *within one meter*:

$$\begin{aligned}\blacksquare \text{ logit } P_M(m_1) - \text{logit } P_M(m_2) &= \alpha \cdot \ln(m_1/m_2) && - \text{ independent of } L^* \\ \blacksquare \text{ logit } P_D(J_1) - \text{logit } P_D(J_2) &= \ln(J_1/J_2) && - \text{ independent of } L^*\end{aligned}$$

**Corollary.** Within a single meter, not only the ranking but the **entire log-odds gap** is scale-free;  $L^*$  fixes only where the  $\frac{1}{2}$ -point sits. (Verified: the electron–proton  $P_M$  log-odds gap is  $-15.0306891424$  at  $L^* = 1 \text{ fm}, 1 \text{ pm}, 1 \text{ nm}, 1 \text{ }\mu\text{m}$  — identical to ten decimals.) ■

### (b) Combined score — invariance FAILS (negative result)

The meters are combined **on the probability scale** ( $P_L = P_D^{(1-\eta)} \cdot P_M^\eta$ , then  $P = \sqrt{P_S \cdot P_L}$ ). Writing  $g(\ell) \equiv \log \sigma(\ell)$ , the combined log-score is  $\log P_L = (1-\eta) \cdot g(\ell_D + 2c) + \eta \cdot g(\ell_M + \alpha c)$  with  $c = \ln \lambda$ . Since  $g$  is **strictly concave**, a translation of its argument does **not** pass through as an additive constant and therefore does **not** cancel in the difference  $\log P_L^A - \log P_L^B$ . The shift survives *inside* a non-linear function, and the ranking can invert.

(v1.5.1 correction: the v1.5 draft attributed the failure to the meters shifting by **different** amounts,  $\alpha \ln \lambda$  vs  $2 \ln \lambda$ . That explanation is **wrong** — the failure occurs even when the shifts are **equal**. The true mechanism is the concavity of  $\log \sigma$ , which re-weights each meter differentially across systems:  $\partial \log P_L / \partial \ell_D = (1-\eta)(1-P_D)$  and  $\partial \log P_L / \partial \ell_M = \eta(1-P_M)$  both depend on  $L^*$  **and** on the system. Asymptotically the aggregation rule even changes character — as  $L^* \rightarrow 0$  it ranks by the arithmetic mean of logits; as  $L^* \rightarrow \infty$  it degenerates to a soft-minimum dominated by the worst meter.)

$$\blacksquare \text{ There exist systems } A, B \text{ and scales } \lambda \text{ with } P(A) > P(B) \text{ at } L^* \text{ but } P(A) < P(B) \text{ at } \lambda L^*.$$

Explicit counterexample **with equal shifts** ( $\alpha = 2$ ,  $\eta = \frac{1}{2}$ ,  $P_S \equiv 1$ ; base log-odds  $A = (\ell_D, \ell_M) = (0, 0)$ ,  $B = (-2, +3)$  — note  $B$  has the *larger* logit mean):

| $L^*$ | $P(A)$   | $P(B)$   | winner   |
|-------|----------|----------|----------|
| 0.01  | 0.010000 | 0.012834 | <b>B</b> |
| 0.10  | 0.099504 | 0.122618 | <b>B</b> |
| 1.00  | 0.707107 | 0.580492 | <b>A</b> |
| 10.0  | 0.995037 | 0.982214 | <b>A</b> |

A pure change of reference scale reverses the winner. A randomized sweep over  $\lambda \in [10^{-3}, 10^3]$  flips **15.3 %** of random pairs at the canonical  $\alpha = 2$ ,  $\eta = \frac{1}{2}$  (29.7 % at  $\alpha = 1$ ; at  $\alpha = 1$  the ordering is not even monotone in  $L^*$  — it can reverse twice). ■

(c) The dichotomy that makes PMD usable

The flips are not arbitrary — they are **exactly** confined to trade-off comparisons:

| comparison type  | definition                                               | $L^*$ -robust?                                                                                                                          | measured flip rate             |
|------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Dominance</b> | one system is $\geq$ the other on <b>every</b> sub-meter | <b>YES — provably 0</b> : the shifts are system-independent and every meter is monotone, so a uniform shift cannot cross the components | <b>0 / 10 093</b>              |
| <b>Trade-off</b> | system A wins some sub-meters and loses others           | <b>NO</b>                                                                                                                               | <b>8 975 / 14 998 (59.8 %)</b> |

■ A PMD ranking is reference-scale-robust  $\Leftrightarrow$  it is a DOMINANCE ranking.

Mandatory methodological consequence (binding on all PMD use).

- 1. **Dominance conclusions** ("system A is better-localized than B on every sub-meter") are objective and may be reported without reference to  $L^*$ .
- 2. **Trade-off conclusions** are defined **only relative to a pre-registered  $L^*$**  and **must** be reported together with an  **$L^*$ -sensitivity analysis** (the range of  $L^*$  over which the ordering is stable). Reporting a trade-off ranking as if it were scale-free is an error.

(d) Why the obvious fix is rejected

Aggregating in log-odds space instead —  $P_L = \sigma((1-\eta) \cdot \text{logit } P_D + \eta \cdot \text{logit } P_M)$  — **does** restore exact invariance (0 / 4 000 flips in every parameter setting tested), because the combined log-odds then shifts uniformly by  $[(1-\eta) \cdot 2 + \eta \cdot \alpha] \cdot \ln \lambda$ . (Confirmed independently:  $\text{Logit } P_L^A - \text{Logit } P_L^B$  is constant to six decimals across  $L^* = 0.01, 1, 100$ . More generally, **any affine rule in logit space with system-independent**

*coefficients preserves ranking, and no non-affine aggregation of the probabilities does.*) It is rejected as canon for a specific, checkable reason: it **destroys the constant elasticity** that connects PMD to **SSS**. Under the log-odds mean,  $\partial \ln U / \partial \ln P_S$  is no longer the fixed  $1/6$  of PMD-7 but becomes *state-dependent* ( $0.2083$  at  $(P_S, P_L) = (0.6, 0.4)$ ;  $0.1667$  at  $(0.9, 0.9)$ ), so the Position pillar would no longer carry a fixed  $1/3$  share of  $U$ .

- Trade-off: geometric aggregation  $\Rightarrow$  SSS-compatible fixed weights,  $L^*$ -dependent trade-off rankings.  
                     log-odds aggregation  $\Rightarrow$  exact scale-invariance, state-dependent weights (breaks PMD-7).

PMD keeps **geometric** aggregation (SSS compatibility is non-negotiable in this corpus) and pays for it with the **explicit** dominance/trade-off discipline of (c) — an honest, quantified cost rather than a hidden one.

## 17. Upgraded massive adapter (Part I §5 $P_L = P_M \rightarrow v1.3+$ proposal) · [L3]

Part I's current  $P_L = P_M$  in the massive sector cannot distinguish a **localized** massive particle from a **massive momentum eigenstate delocalized over all space**. The proposed fix folds in the detection channel:

- massive sector:  $P_L = P_D^{(1-\eta)} \cdot P_M^\eta$ ,  $0 \leq \eta \leq 1$
- massless sector:  $P_L = P_D$

At the symmetric point  $\eta = \frac{1}{2}$ ,  $P_L = \sqrt{P_D \cdot P_M}$ . The two essential negative controls (Part I §8) are then satisfied **by construction**:

- massive-but-delocalized :  $P_D = 0, P_M > 0 \Rightarrow P_L = 0$  (correctly not localized)
- massless-but-detected :  $m = 0, P_D > 0 \Rightarrow P_L = P_D > 0$  (correctly not zero-Position)

This is strictly stronger than the  $P_L = P_M$  branch because it embeds the two controls into the formula rather than relying on a case split. **Status:** adopted into the Part I §5 core meter in **v1.4**;  $\eta$  must be **pre-registered**, never fit post hoc. **Two v1.5 caveats now attach to this form:** (i) it **double-counts mass**, because  $J_C$  behind  $P_D$  is itself bounded by  $m$  (§15.1 / PMD-10) — the non-redundant  $P_{cap} / P_{eff}$  blend is the recommended calibration rival; (ii) trade-off rankings produced by this blend are **not** reference-scale invariant (§16.1 / PMD-11b) and require an  $L^*$ -sensitivity report.

## 18. Spatial-support meter $P_S$ · [L3: modelling choice]

Let  $p(x)$  be a normalized spatial distribution over a pre-declared reference region  $\Omega$ , and  $u_\Omega = 1/|\Omega|$  the uniform. Define the distinguishability information as a KL contrast:

$$D_S = D_{KL}(p \parallel u_\Omega) = \int_\Omega p(x) \cdot \ln[ p(x) / u_\Omega(x) ] dx \geq 0 \quad (\text{Gibbs; } =0 \text{ iff } p=u_\Omega \text{ a.e.})$$

and

$$\blacksquare \quad P_S = 1 - e^{(-D_S / D^*)}, \quad D^* > 0$$

so  $P_S \in [0,1)$ ,  $P_S = 0 \Leftrightarrow p = u_\Omega$  (no distinguishable "where"),  $\partial P_S / \partial D_S = (1/D^*)e^{(-D_S/D^*)} > 0$ . This operationalizes "constrained support" as an information contrast against a chosen context.

## 19. Theorem PMD-6 — uniqueness of the nested geometric mean · [L3: conditional]

Assume the Position aggregator is **multiplicatively separable**,  $G(P_S, P_L) = f(P_S) \cdot f(P_L)$ , and require symmetry, idempotence  $G(t, t) = t$ , zero-annihilation, and monotonicity. Idempotence gives  $f(t)^2 = t$ , so  $f(t) = \sqrt{t}$ , hence

$$\blacksquare \quad G(P_S, P_L) = \sqrt{P_S \cdot P_L}$$

The geometric nesting is therefore **not arbitrary** — it is forced once separability + symmetry + idempotence + non-compensation are accepted. ■

**Scope of the uniqueness (honest).** This is uniqueness *within the axiom set*. Rival aggregators that violate one axiom — the minimum  $\min(P_S, P_L)$ , a soft-min, or a weighted geometric mean  $P_S^a \cdot P_L^b$  with  $a \neq b$  (breaks symmetry) — are **not refuted**; they simply fall outside {multiplicative separability, symmetry, idempotence}. CEPT **CORE.PMD.08** records this as *conditionally proved*, and the four-model comparison of §29.9 ( $P_S$  ;  $P_S \cdot P_D$  ;  $P_S \cdot P_M$  ;  $P_S \cdot P_D \cdot P_M$ ) is the model-comparison arm where such rivals are tested.

## 20. Theorem PMD-7 — full nested U and elasticity (no double-weighting) · [L2 given the meters]

With  $P = \sqrt{P_S \cdot P_L}$  and  $U = \sqrt[3]{F \cdot P \cdot A}$ :



$$\blacksquare U = F^{(1/3)} \cdot A^{(1/3)} \cdot P_S^{(1/6)} \cdot P_L^{(1/6)}$$

Log-elasticities:

$$\begin{aligned} \partial \ln U / \partial \ln F &= 1/3 & \partial \ln U / \partial \ln A &= 1/3 \\ \partial \ln U / \partial \ln P_S &= 1/6 & \partial \ln U / \partial \ln P_L &= 1/6 \end{aligned}$$

The two Position sub-meters together carry  $1/6 + 1/6 = 1/3$  — **exactly** Position's original share. The nesting **splits** Position's existing weight into two equal halves; it does **not** add weight. (With the §17 adapter  $P_L = P_D^{(1-\eta)} \cdot P_M^\eta$ , the  $P_L$  share  $1/6$  splits further into  $P_D:(1-\eta)/6$  and  $P_M:\eta/6$  — at  $\eta=1/2$ ,  $1/12$  each; see §21.) This is what keeps PMD compatible with **SSS**, where the three top-level pillars aggregate geometrically.

## 21. Full proposed PMD operational formula

**Massive sector** ( $\eta$ -weighted adapter):

$$\begin{aligned} P_M &= (mcL^*/\hbar)^\alpha / (1 + (mcL^*/\hbar)^\alpha) \\ P_D &= L^{*2} \cdot J_C / (1 + L^{*2} \cdot J_C) \\ P_L &= P_D^{(1-\eta)} \cdot P_M^\eta \\ P &= \sqrt{P_S \cdot P_L} \\ \blacksquare U_{PMD} &= F^{(1/3)} \cdot A^{(1/3)} \cdot P_S^{(1/6)} \cdot P_D^{((1-\eta)/6)} \cdot P_M^{(\eta/6)} \end{aligned}$$

**Massless sector** ( $P_L = P_D$ ; there is **no**  $P_M = 0$ , only an *inapplicable* mass channel):

$$\blacksquare U_{PMD}^{(0)} = F^{(1/3)} \cdot A^{(1/3)} \cdot P_S^{(1/6)} \cdot P_D^{(1/6)}$$

## 22. Theorem PMD-8 — interactions participate in composite invariant mass · [L1: standard, with signed correction]

For a closed system with total four-momentum  $P^\mu$ :

$$\blacksquare M^2 c^2 = P_\mu P^\mu, \quad \text{and in the centre-of-momentum frame } (P=0): \quad \blacksquare M c^2 = E_{\text{COM}}$$

Writing  $H = \sum_i m_i c^2 + H_{\text{kin,int}} + H_{\text{field}} + H_{\text{int}}$ :

$$\blacksquare M - \sum_i m_i = ( \langle H_{\text{kin,int}} \rangle + \langle H_{\text{field}} \rangle + \langle H_{\text{int}} \rangle ) / c^2$$

This is the exact, rigorous sense in which a system's **internal relations participate in its invariant mass**.

**Essential sign correction (honest).** The sign is **not** universally positive:

- **Nucleons:** QCD dynamics contribute a *large positive* mass over the sum of current-quark masses.
- **Atomic / nuclear bound states:** binding energy gives a *mass defect* (mass **below** the sum of separated parts).

So the defensible statement is **interaction context MODIFIES invariant mass**, **not more context always means more mass**. Define a **signed** interaction-mass fraction:

```

χ_int = (M - Σ_i m_i) / M
χ_int > 0 : interactions raise total mass over the constituent baseline
(e.g. QCD)
χ_int < 0 : binding mass defect (e.g. nuclei, atoms)
χ_int = 0 : no net interaction contribution vs baseline

```

**χ\_int** is a real candidate for a *partial* operationalization of "internal relational context."

**Why the composite (QCD) case still strengthens the reading** (see Part I §8): a nucleon's mass is the energy of its **own internal interactions with itself** made inertial — the "mass = crystallized context" reading applied to a system's relations with its own parts. The signed correction simply forbids the *slogan* "more context ⇒ more mass" while keeping the *structural* claim "interaction context is inertial."

## 23. Theorem PMD-9 — no-go for automatic empirical surplus · [L1: information theory — the honesty gate]

Let the standard model of the phenomenon use all standard variables  $X_{std} = (m, p^\mu, p, H_{int}, C_{measurement}, \dots)$ , and let the PMD "context" variable be a **deterministic function** of them,  $C = f(X_{std})$ . Then for **every** observable  $Y$ :

$$\blacksquare I(Y ; C \mid X_{std}) = 0$$

**Proof.**  $C = f(X_{std}) \Rightarrow H(C \mid X_{std}) = 0$ , so  $I(Y;C|X_{std}) = H(C|X_{std}) - H(C|Y,X_{std}) = 0 - 0 = 0$ . ■

**Consequence (the discipline).** If "context" is merely a **relabelling** of variables standard physics already uses, PMD **cannot** add predictive information. Empirical surplus **requires** an *independently measured* context observable  $C_R$  with

$$\blacksquare I(Y ; C_R \mid X_{std}) > 0$$

This is the formal version of the Part I §8 null model  $H_0$ . PMD is honest precisely because it proves its own emptiness absent a new observable.

## 23.1. Theorem PMD-12 — the exact surplus identity · [L1: proved]

(New in v1.5. Turns §24's  $\Delta_{CV}$  from an ad hoc score into an estimator of a specific information quantity, and makes PMD-9 its zero case.)

For the **true** conditional distributions, the expected log-loss (in nats) of the null model is  $\mathbb{E}[\ell_0] = H(Y | X_{std})$  and of the PMD model  $\mathbb{E}[\ell_1] = H(Y | X_{std}, C_R)$ . Hence:

$$\blacksquare \quad \mathbb{E}[\ell_0] - \mathbb{E}[\ell_1] = H(Y|X_{std}) - H(Y|X_{std}, C_R) = I(Y ; C_R | X_{std})$$

**Proof.** Both terms are conditional entropies of the same  $Y$ ; their difference is the conditional mutual information by definition.  $\blacksquare$  (Valid for differential entropies too — the individual  $h(\cdot)$  are not reparametrization-invariant but their difference is. For continuous  $C$  the no-go should be argued from conditional independence —  $C$  is a.s. constant given  $X_{std}$  — not from  $I \leq H(C|X_{std}) = 0$ , since  $h(C|X_{std}) = -\infty$  there.) (Verified numerically on a random joint distribution:  $I = 0.095060227113$  versus  $H(Y|X) - H(Y|X, C) = 0.095060227113$ ; and for a deterministic  $C = f(X)$ ,  $I = 0.00e+00$  — PMD-9 recovered exactly.)

### Three consequences that sharpen the empirical programme:

1. **PMD-9 is the zero case.**  $C = f(X_{std}) \Rightarrow I(Y; C|X_{std}) = 0 \Rightarrow$  the achievable log-loss gain is *exactly* zero, not merely "unproven." The no-go and the surplus criterion are one theorem seen from two sides.
2. **The effect size has units.** Any claimed PMD improvement is quantified in **nats per observation**, so pre-registration can name a *numerical* threshold ( $\Delta \geq \delta$  nats with a CI excluding 0) instead of a vague "beats the null."
3. **Held-out loss is only a proxy estimator — with four documented failure modes (v1.5.1).** The identity holds for the **true** conditionals;  $\Delta_{CV}$  inherits none of that automatically:
  - **Misspecification breaks the bridge (the important one).** Under a *restricted* model class, a deterministic  $C = f(X_{std})$  with  $I(Y; C|X_{std}) = 0$  **can still strictly improve held-out loss** — handing a non-linear feature to a linear/logistic model is the canonical case. A positive  $\Delta_{CV}$  is therefore **not** evidence that  $I(Y; C_R|X_{std}) > 0$ ; it may only measure the model class's inability to represent  $f$ .
  - **$\Delta_{CV}$  is biased downward.** k-fold CV estimates the risk of the *fitting procedure*; the two model classes differ in complexity, so their optimism terms do not cancel and the larger model is penalized. Hence  $\Delta_{CV} \leq 0$  does **not** refute  $I > 0$ .
  - **Fold-wise inference is unreliable.** CV folds are dependent and there is no unbiased variance estimator for k-fold CV; naive t-tests across folds are anticonservative. Use a **permutation null on  $C_R$**  with the entire pipeline inside the permutation.

- **Plug-in mutual-information estimates are biased upward**, so  $\hat{I} > 0$  on finite data is expected even when  $I = 0$ ; a null distribution is required, not a point estimate.
4. **Scope of the no-go.**  $C = f(X_{std})$  must be a function of **exactly the conditioning set**. If  $C = f(X_{full})$  while only  $X_{std} \subset X_{full}$  is conditioned on, then  $I(Y; C | X_{std})$  is generically  $> 0$ . Note also that  $I(Y; C | X) = 0$  neither implies nor is implied by  $I(Y; C) = 0$  (explaining-away): a marginally useless  $C_R$  can be conditionally informative, and vice versa.

## 24. Pre-registered PMD empirical model (how surplus would be earned)

**Outcome must be a raw observable, never the constructed  $P_D$** . Using  $P_D$  as the target would be **circular** — it is built from the same channel  $C$  as the predictors. Let  $Y$  be a directly measured quantity: the empirical localization error  $Var(\hat{x})$ , a detection hit-rate in region  $R$ , or a resolved position residual.

**Null model.**  $H_0: g(Y) = f_{std}(m, \Delta p, E, POVM, geometry) + \epsilon$

**PMD model.**

$$H_1: g(Y) = \beta_0 + \beta_m \cdot \ln x_M + \beta_C \cdot C_R + \beta_{\{mC\}} \cdot (\ln x_M) \cdot C_R + \epsilon$$

where  $g$  is a fixed link (e.g.  $\ln Var(\hat{x})$ ) and  $C_R$  is an **independently measured** relational-context variable (**not** a deterministic function of  $X_{std}$  — otherwise PMD-9 forces zero surplus).

**PMD earns content only if**, on held-out data,  $\beta_C \neq 0$  or  $\beta_{\{mC\}} \neq 0$ , and

$$\Delta_{CV} = \text{Loss}(H_0) - \text{Loss}(H_1) > 0 \quad (\text{pre-set statistical margin}),$$

equivalently  $I(Y; C_R | X_{std}) > 0$ .

**RH** requires exactly this contest against the null and against rivals — not mere internal mathematical coherence.

## 25. Worked example for $P_M$ (illustration only)

With  $\alpha = 2$ ,  $L^* = 1 \text{ fm}$ , and  $P_M = (L^*/\lambda_C)^2 / (1 + (L^*/\lambda_C)^2)$ :

| particle          | $\lambda_C$                             | $L^*/\lambda_C$ | P_M                                                                                                       |
|-------------------|-----------------------------------------|-----------------|-----------------------------------------------------------------------------------------------------------|
| electron          | 386.159 fm                              | 0.00259         | $6.7 \times 10^{-6}$                                                                                      |
| proton            | 0.2103 fm                               | 4.755           | 0.958                                                                                                     |
| photon (massless) | —<br>( $\lambda_C \rightarrow \infty$ ) | —               | <b>N/A</b> — mass channel inapplicable; scored via $P_D > 0$ (§17 massless adapter), <b>not</b> $P_L = 0$ |

This does **not** mean the proton "has Position 0.958" and the electron "has almost none." It means only: *relative to the task scale  $L^* = 1\text{ fm}$ , the proton's mass Compton-capacity proxy is near-saturated and the electron's is not.* Choose a different  $L^*$  and the numbers change. Therefore  $L^*$  must be **task-specific, pre-registered, and never chosen after seeing the result.** (All numbers verified numerically.)

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## 26. Claim–Evidence–Prediction–Test map

| ID           | Claim                                                                   | Status                                                                                                                                      | Basis                               | Test                             |
|--------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------|
| CORE.PMD.01  | $m \cdot \lambda_C = \hbar/c$                                           | established                                                                                                                                 | identity                            | unit check                       |
| CORE.PMD.02  | mass $\leftrightarrow$<br>acceleration-<br>susceptibility<br>reciprocal | established (NR)                                                                                                                            | $a=F/m$                             | fixed-force<br>test              |
| CORE.PMD.03  | $F_Q \leq 8mK/\hbar^2$                                                  | proved (scoped)                                                                                                                             | QFI + kinetic<br>energy             | displacement<br>estimation       |
| CORE.PMD.04  | context channels<br>add Fisher info                                     | proved                                                                                                                                      | conditional<br>independence         | multi-<br>detector<br>experiment |
| CORE.PMD.05  | coarse-graining<br>lowers position info                                 | proved                                                                                                                                      | data processing                     | sensor<br>removal                |
| CORE.PMD.06  | $g_M =$<br>$x^\alpha/(1+x^\alpha)$                                      | conditionally<br>proved                                                                                                                     | log-odds axiom                      | calibration                      |
| CORE.PMD.07  | $\alpha = 2$                                                            | model choice                                                                                                                                | inverse-variance<br>scaling         | model<br>comparison              |
| CORE.PMD.08  | $P = \sqrt{P_S \cdot P_L}$                                              | conditionally<br>proved                                                                                                                     | separability +<br>symmetry          | rival<br>aggregators             |
| CORE.PMD.09  | interactions modify<br>composite mass                                   | established                                                                                                                                 | invariant mass                      | mass-defect<br>data              |
| CORE.PMD.10  | context <i>always</i><br>increases mass                                 | <b>false in general</b>                                                                                                                     | sign varies ( $\chi_{\text{int}}$ ) | binding<br>controls              |
| CORE.PMD.11  | mass = context                                                          | L3 interpretation                                                                                                                           | no derivation                       | independent<br>$C_R$             |
| CORE.PMD.12  | PMD adds<br>prediction                                                  | <b>not yet</b>                                                                                                                              | requires $I>0$                      | held-out test                    |
| CORE.PMD.13  | $F_Q \leq 4(\langle \hat{H}^2 \rangle$<br>$-m^2c^4)/(\hbar^2c^2)$       | proved ( <b>1 particle,</b><br><b>free, <math>E&gt;0</math>, <math>\langle \hat{H}^2 \rangle</math></b><br><b><math>&lt;\infty</math></b> ) | operator identity<br>+ QFI          | relativistic<br>metrology        |
| CORE.PMD.13b | $F_Q \leq 8mK/\hbar^2$<br>with <b>relativistic</b> $K$                  | <b>FALSE</b> — violated<br>by<br>$(\sqrt{(1+u^2)}+1)/2$<br>at every $u>0$                                                                   | §12.1<br>counterexample             | fixed- $K$<br>metrology          |
| CORE.PMD.14  | "more mass $\Rightarrow$<br>more capacity"                              | <b>only at fixed</b><br><b>kinetic energy</b>                                                                                               | reverses at fixed<br>total $E$      | budget must<br>be declared       |
| CORE.PMD.15  | $J_C = \varepsilon \cdot u \cdot (8mK/\hbar^2)$                         | <b>tautology</b><br>(definition);                                                                                                           | Braunstein–Caves<br>+ PMD-2         | efficiency<br>estimation         |

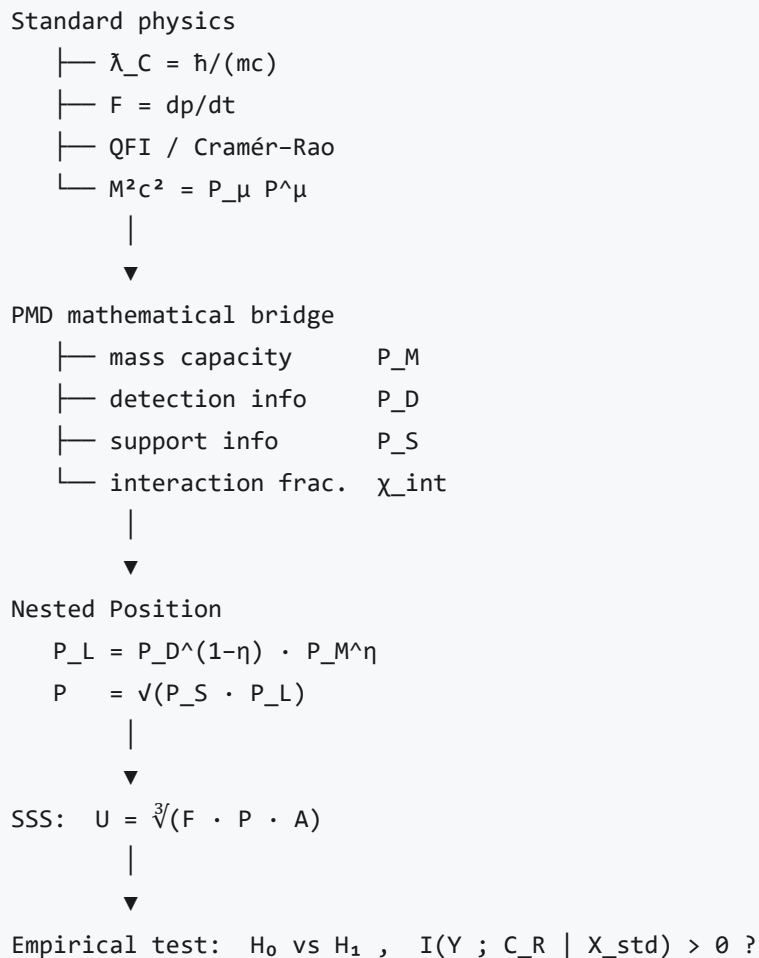
| ID          | Claim                                                           | Status                                                                                         | Basis                                                 | Test                                                              |
|-------------|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------------------|
|             |                                                                 | content is $\epsilon, u \in [0,1]$ , NR-scoped                                                 |                                                       |                                                                   |
| CORE.PMD.16 | $P_D^{(1-\eta)}P_M^\eta$<br>double-counts<br>mass               | <b>WITHDRAWN</b><br><b>v1.5.1</b> — bound is<br>one-directional;<br>dependence is<br>empirical | replaced by<br>$\eta_{\text{eff}} = \eta + (1-\eta)s$ | estimate $s$<br>on corpus                                         |
| CORE.PMD.17 | per-meter ranking<br>is $L^*$ -invariant                        | proved                                                                                         | uniform log-<br>odds shift                            | scale sweep                                                       |
| CORE.PMD.18 | combined-score<br>ranking is $L^*$ -<br>invariant               | <b>FALSE</b>                                                                                   | counterexample;<br>59.8 % of trade-<br>off pairs flip | dominance-<br>only rule                                           |
| CORE.PMD.19 | $\mathbb{E}[\Delta \log\text{-loss}] = I(Y;C_R X_{\text{std}})$ | proved <b>for true<br/>conditionals only</b>                                                   | conditional<br>entropies                              | permutation<br>null; $\Delta_{\text{CV}}$ is<br>a biased<br>proxy |

## 27. Variable & equation registry

| symbol                                    | meaning                                                            | units             |
|-------------------------------------------|--------------------------------------------------------------------|-------------------|
| $m$                                       | invariant mass                                                     | kg                |
| $\tilde{\lambda}_C$                       | reduced Compton wavelength                                         | m                 |
| $L^*$                                     | target localization scale                                          | m                 |
| $m^* = \hbar/(cL^*)$                      | reference mass                                                     | kg                |
| $x_M = m/m^*$                             | Compton ratio                                                      | —                 |
| $J_C$                                     | classical Fisher information for position                          | $m^{-2}$          |
| $F_Q$                                     | quantum Fisher information                                         | $m^{-2}$          |
| $D_S$                                     | KL support information                                             | —                 |
| $P_S$                                     | support / distinguishability score                                 | [0,1)             |
| $P_D$                                     | detection / localization score                                     | [0,1)             |
| $P_M$                                     | massive Compton-capacity score                                     | [0,1)             |
| $P_L$                                     | combined localizability score                                      | [0,1)             |
| $\eta$                                    | mass-proxy weight                                                  | —                 |
| $\chi_{\text{int}}$                       | signed interaction-mass fraction                                   | —                 |
| $C_R$                                     | independently measured context                                     | protocol-specific |
| $\epsilon = J_C/F_Q$                      | context efficiency (extraction fraction)                           | [0,1]             |
| $u = F_Q/(8mK/\hbar^2)$                   | state utilization of the mass-energy ceiling                       | [0,1]             |
| $E_{\text{kin}} = \hat{H} - mc^2$         | relativistic kinetic energy                                        | J                 |
| $(\Delta H)^2$                            | energy variance                                                    | $J^2$             |
| $\lambda$                                 | reference-scale rescaling factor ( $L^* \rightarrow \lambda L^*$ ) | —                 |
| $u = p/(mc)$                              | momentum in units of $mc$ (relativity parameter)                   | —                 |
| $s = \partial \log P_D / \partial \log m$ | population elasticity of detection w.r.t. mass                     | [0,1]             |
| $\eta_{\text{eff}} = \eta + (1-\eta)s$    | <b>realized</b> mass weight in the adapter                         | $[\eta, 1]$       |
| $U$                                       | nested stability score                                             | [0,1]             |



## 28. Dependency graph



## 29. Open issues / calibration tasks (v1.5 roadmap)

1. Corrected §5 wording (done — v1.2.1).
2. Replace  $P_L = P_M$  with the  $P_D$ -inclusive adapter — **done in v1.4** (§5 core now  $P_L = P_D^{(1-\eta)} \cdot P_M^\eta$ ,  $\eta=1/2$  default;  $P_L=P_M$  kept as  $\eta=1$  case).
3. Choose or compare  $\alpha \in \{1, 2, \text{free}\}$ .
4. Fix  $\eta$  by pre-registration, never post hoc.
5. Declare the reference region  $\Omega$  for  $P_S$ .
6. Set  $L^*$  from task resolution, not from the result.
7. Separate internal-context from external/Machian context.
8. Use the **signed**  $\chi_{int}$  (interactions can lower mass).
9. Compare four models:  $P_S$  ;  $P_S \cdot P_D$  ;  $P_S \cdot P_M$  ;  $P_S \cdot P_D \cdot P_M$ .
10. Count PMD as empirically upgraded **only** if  $I(Y ; C_R \mid X_{std}) > 0$  — now quantified in **nats** by PMD-12 (§23.1).
11. **Resolve the mass double-count** (§15.1): calibrate  $P_D^{(1-\eta)} P_M^\eta$  against the non-redundant  $P_{cap}^{(1-\eta)} \cdot P_{eff}^\eta$ .
12. **Report  $L^*$ -sensitivity** for every trade-off ranking (§16.1c) — dominance rankings are exempt.

13. **Declare the energy budget** (fixed kinetic vs fixed total-  $\langle \hat{H}^2 \rangle$  ) in every capacity claim (§12.1) — the mass effect reverses between them.
14. **Estimate**  $s$  (and hence  $\eta_{\text{eff}}$  ) on the scored corpus (§15.1) —  $\eta$  alone does not determine the realized mass weight.
15. **Use a permutation null** for any surplus claim, and never read  $\Delta_{\text{CV}} > 0$  as  $I > 0$  without a well-specified model class (§23.1).

## 30. Strongest formulation of the hypothesis (replaces "mass = concentrated context" as the core)

**PMD Localization-Leverage Hypothesis.** In the massive non-relativistic sector, invariant mass bounds the spatial Fisher information available per unit kinetic-energy budget,

$$F_Q / K \leq 8m / \hbar^2$$

and the measurement context bounds how much of that available information is actually extracted,

$$J_C \leq F_Q .$$

Hence mass and context have distinct, complementary roles:

■ mass sets capacity ; context realizes accessibility .

**Scope (v1.5).** "Capacity" is defined at a **fixed kinetic-energy budget**; at fixed *total* energy the mass dependence reverses (§12.1). The three-way split of that capacity is exact:  $J_C = \epsilon \cdot u \cdot (8mK/\hbar^2)$  (§15.1).

So the defensible corpus statement is **not**  $\text{Mass} = \text{Context}$  but

■ Position quality = f( massive capacity , spatial state , measurement context )

## References & provenance

- **Parent record:** U-Theory / U-Model — DOI **10.17605/OSF.IO/74XGR** · <https://u-model.org>
- **Sibling appendices:**  $\text{MMT}$  (matter = crystallized meaning),  $\text{DIM}$  ,  $\text{ST}$  /DPR (currency registry),  $\text{QMC}$  (measurement/localization),  $\text{GEN}$  ,  $\text{SSS}$  ,  $\text{RH}$  ,  $\text{POS}$  .

- **Physics anchors (established):** the **reduced** Compton wavelength  $\lambda_C = \hbar/mc$  (ordinary  $\lambda_C = h/mc$ ) & the pair-creation one-particle localization limit; the relativistic dispersion  $E^2 = p^2 c^2 + m^2 c^4$  and photon  $p = E/c$ ; inertia via  $F = dp/dt$  ( $F \approx ma$  at low  $v$ ); the absence of a sharp Newton–Wigner position operator for massless spin  $> 1/2$ ; quantum Fisher information & the Cramér–Rao bound; Fisher-information additivity and the data-processing inequality; invariant mass  $M^2 c^2 = P_\mu P^\mu$ ; Mach's principle (historical, only partially realized in GR via frame-dragging); the Higgs mechanism (elementary masses; composite mass mostly QCD binding); the position–momentum uncertainty relation.
  - **Author:** Petar Nikolov (ORCID 0009-0001-8669-2276). © 2026, CC BY 4.0 (text) / MIT (any code).
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## Changelog

| version | date       | change                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v1.4.1  | 2026-07-25 | Final review polish: $P_L$ registry range $\rightarrow [0, 1]$ (consistent with $P_S / P_D / P_M$ ); §8 "as of" version bumped to v1.4; §19 cross-reference rewired to §29.9 four-model comparison (§24 has no explicit aggregator arm).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| v1.5.1  | 2026-07-25 | <b>Self-correction after adversarial review of the v1.5 draft — three real errors fixed, one claim withdrawn.</b> (1) <b>PMD-2R inference was backwards:</b> the non-negative relativistic corrections mean the relativistic ceiling <i>exceeds</i> $8mK/\hbar^2$ , so PMD-2 is <b>not</b> a relativistic bound — it is violated at <i>every</i> nonzero momentum by the exact factor $(\sqrt{(1+u^2)}+1)/2$ (1.207 at $u=1$ , 50.5 at $u=100$ ). PMD-2 is now stated as valid <b>iff</b> $K \equiv \langle \hat{p}^2 \rangle / 2m$ , i.e. as a rewriting, and strictly NR. (2) $E^2 \rightarrow \langle \hat{H}^2 \rangle$ : at fixed <i>mean</i> energy the ceiling is unbounded ( $(\Delta H)^2$ is free), so the total-energy budget must be stated as fixed second moment; the reversal was also mislabelled relativistic-vs-NR when it is fixed- $K$ -vs-fixed- $E$ . (3) <b>PMD-11's mechanism was wrong:</b> the failure is caused by the <b>concavity of <math>\log \sigma</math></b> , not by unequal logit shifts — it occurs even at $\alpha=2$ where the shifts are equal (new explicit counterexample table). (4) <b>Withdrawn:</b> the "adapter double-counts mass" claim — the bound is one-directional, $J_C$ is not monotone in $m$ , and dependence is empirical; replaced by the checkable $\eta_{\text{eff}} = \eta + (1-\eta)s$ . Also: PMD-2R scoped to one-particle/free/positive-energy/ $\langle \hat{H}^2 \rangle < \infty$ (the identity <b>fails for <math>N&gt;1</math></b> ); PMD-10 demoted from theorem to a definitional decomposition with $\epsilon, u$ range scopes; PMD-12 given four estimator caveats (misspecification, CV bias, fold dependence, plug-in MI bias). |
| v1.5    | 2026-07-25 | <b>Mathematical-apparatus upgrade (4 new theorems, 1 negative result).</b> <b>PMD-2R</b> (§12.1): exact relativistic bound $F_Q \leq 4(\langle \hat{H}^2 \rangle - m^2 c^4) / (\hbar^2 c^2)$ via the operator identity $\hat{H}^2 = \hat{p}^2 c^2 + m^2 c^4$ , with PMD-2 recovered as its leading term — closes the NR scope gap; <b>plus the parametrization-reversal caveat</b> (mass raises the ceiling at fixed <i>kinetic</i> energy, lowers it at fixed <i>total</i> energy), and <b>one concrete falsifiable ceiling</b> $\text{Var}(\hat{x}) \geq \hbar^2 / (8NmK)$ . <b>PMD-10</b> (§15.1): capacity–efficiency factorization $J_C = \epsilon \cdot u \cdot (8mK/\hbar^2)$ , which <b>proves a mass double-count</b> in the $P_D^\wedge(1-\eta)P_M^\wedge\eta$ adapter and supplies the non-redundant $P_{\text{cap}} / P_{\text{eff}}$ alternative. <b>PMD-11</b> (§16.1): per-meter reference-scale invariance <b>proved</b> , combined-score invariance <b>disproved</b> with an explicit counterexample (59.8 % of trade-off pairs flip), yielding the binding <b>dominance / trade-off</b> rule and an $L^*$ -sensitivity requirement; the log-odds fix is examined and rejected because it breaks PMD-7's fixed $1/6$ elasticity (SSS compatibility). <b>PMD-12</b> (§23.1): exact surplus identity $\mathbb{E}[\Delta \log\text{-loss}] = I(Y; C_R   X_{\text{std}})$ , making PMD-9 its zero case and giving effect sizes in nats. All results verified numerically. <b>No new physics.</b>                                                                                                                                                                                                |
| v1.4    | 2026-07-25 | <b>Review-hardening (P0/P1/P2).</b> Adopted the §17 adapter into the §5 core meter: $P_L = P_D^\wedge(1-\eta) \cdot P_M^\wedge\eta$ (pre-registered $\eta=\frac{1}{2}$ ; $P_L=P_M$ retained as the $\eta=1$ legacy Compton reading) — the two negative controls now hold <i>by construction</i> , resolving the §5 $\leftrightarrow$ §17 dual-state. Clarified context $\coloneqq$ <b>channel <math>C</math></b> vs its Fisher <b>content <math>J_C</math></b> (§5). Added: PMD-2 non-relativistic <b>scope</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

| version       | date       | change                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               |            | <b>caveat</b> (§12); PMD-4 <b>regularity</b> note (§14); PMD-6 <b>rival-aggregator</b> scope (§19); $[0,1]$ ranges (§10, §27); <b>raw-observable</b> protocol to kill circularity in §24 ( <b>Y</b> must not be <b>P_D</b> ); a massless-photon <b>negative-control row</b> (§25); PMD-7 adapter elasticity split (§20); epigraph <i>figurative</i> footnote; minimal reading-path guide; §29 renamed to v1.5 roadmap. <b>No new physics; PMD-2/PMD-9/<math>\chi_{int}</math>/§30 untouched.</b>                                            |
| <b>v1.3</b>   | 2026-07-25 | <b>Single-document merge.</b> Folded the entire formal layer (former <b>APPENDIX_PMD_MATH</b> ) into this file as <b>PART II (§9–§30)</b> : all theorems PMD-1...9 with proofs, the operational meters ( <b>P_D</b> , <b>P_M</b> , <b>P_S</b> ), the upgraded adapter, the composite-mass $\chi_{int}$ , the empirical no-go, the CEPT map, registry, and dependency graph — nothing dropped. Part I (§0–§8) is the interpretation; per-result epistemic labels live in Part II. Internal cross-references rewired to the merged numbering. |
| <b>v1.2.1</b> | 2026-07-25 | Consultant v1.2 + PMD-MATH §2.3 fix: the residual §5 "massless⇒delocalized" wording corrected (at $m \rightarrow 0$ the mass channel <b>P_M</b> vanishes but <b>P_L</b> via <b>P_D</b> need not; unpaid Position needs <b>both</b> $P_S \rightarrow 0$ and $P_L \rightarrow 0$ ).                                                                                                                                                                                                                                                           |
| <b>v1.2</b>   | 2026-07-25 | Consultant review-hardening: anti-fourth-pillar note (§3); operational <b>Context_proxy</b> placeholder (§5); <b>resist <math>\Delta</math>momentum</b> table wording; QCD internal-interaction justification (§8); Machian scope qualifier (§7); explicit <b>no present empirical surplus</b> clause; changelog table.                                                                                                                                                                                                                     |
| <b>v1.1.2</b> | 2026-07-25 | Review polish draft: QCD strengthens-the-reading argument; §3 nested-sub-price note; §7 Machian qualifier.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>v1.1.1</b> | 2026-07-19 | §8 discipline through the body: massless $\neq$ delocalized; inertia resists $\Delta$ momentum not displacement; Higgs = VEV coupling not drag; massless adapter $g_0$ ; $H_0$ and negative controls.                                                                                                                                                                                                                                                                                                                                       |
| <b>v1.1</b>   | 2026-07-19 | Physics-hardened core: reduced Compton $\lambda_C$ ; relativistic $F = dp/dt$ , $p = \gamma m v$ , photon $p = E/c$ ; nested meter $P = \sqrt{P_S \cdot P_L}$ ; mass <b>P_M</b> as one proxy only in the massive rest-frame sector.                                                                                                                                                                                                                                                                                                         |

Status: **v1.5.1 single document** — Part I is L3/L4 speculative interpretation on a rigorous physical core; Part II is the formal layer (L1/L2 proved core:  $m \cdot \lambda_C = \hbar/c$  ;  $F_Q \leq 8mK/\hbar^2$  ; Fisher additivity  $J_C = \sum J_k$  ; data-processing monotonicity; nested-mean uniqueness  $P = \sqrt{P_S \cdot P_L}$  ; composite mass  $Mc^2 = E_{COM}$  with signed  $\chi_{int}$  ; empirical no-go  $C = f(X_{std}) \Rightarrow I(Y;C|X_{std}) = 0$  ; **v1.5 additions**: exact relativistic bound PMD-2R, capacity–efficiency factorization PMD-10, reference-scale invariance PMD-11 (per-meter **yes**, combined score **no** — dominance rankings only), surplus identity PMD-12 (true conditionals only); **v1.5.1 records three corrected errors and one withdrawn claim from the v1.5 draft**; L3 modelling choices: **P\_M** , **P\_D** , **P\_S** ,  $\alpha = 2$  ,  $\eta$  ). The one-particle reduced-Compton localization scale is textbook physics; reading mass as the Position-currency's context/inertia

*price is a U-Theory interpretation, not a derivation; "mass = context" is **not** upgraded by the formal layer and remains L3/L4. Position pays twice — in space, and in mass — **for a massive particle in its rest frame.***